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ROLL NO : 52

DATE : 10/9/2025

COMPLETED THE PROJECT NAMED AS PHASE 1

TECHNOLOGY PROJECT NAME : USER REGISTRATION WITH VALIDATION

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User Registration with Validation

1. Problem Statement

In today's digital era, most applications, whether mobile or web-based, require users to create an account before accessing personalized services. A user registration system is therefore a fundamental component of any platform that manages individual user identities, provides secure access, and stores sensitive data.

However, designing a registration system without proper validation can lead to major challenges such as:

- Invalid or incomplete data entry (e.g., emails without "@" or missing usernames).
- Duplicate accounts using the same email or username.
- Weak passwords, which may compromise account security.
- Poor user experience caused by unclear error messages or confusing registration steps.

Problem Summary:

The goal is to develop a User Registration with Validation system that allows new users to create accounts securely and reliably. This system must validate inputs such as username, email, and password, enforce uniqueness, and ensure data security while providing users with clear, actionable feedback for invalid entries.

2. Users and Stakeholders

1. End Users:

- o Individuals signing up to access the system.
- o Need an easy-to-use and secure registration process.

2. System Administrators:

- o Manage user accounts and ensure proper functioning of the system.
- o Responsible for troubleshooting issues like failed registrations or account duplication.

3. Business Owners / Stakeholders:

- o Interested in smooth onboarding of users, reduced drop-off rates during registration, and secure handling of personal data.

4. Developers:

- o Responsible for building the registration system with proper validation logic.
- o Ensure that the process is user-friendly, scalable, and secure.

3. User Stories

1. As a new user, I want to register with a valid email and password so that I can create an account and access the system.

2. As a user, I want to be notified if my email or username already exists so that I can choose a different one.

3. As a user, I want to be guided with clear error messages if I provide invalid inputs (like weak password, invalid email format, or mismatched confirm password).

4. As a system administrator, I want the registration system to enforce strong password policies so that user accounts remain protected.

5. As a developer, I want standardized validation rules so that all accounts follow the same requirements, ensuring consistency and reducing errors.

4. MVP Features

For the Minimum Viable Product (MVP), the following features are essential:

1. Registration Form Fields:

- o Username → Required, minimum 3 characters, alphanumeric only.
- o Email → Required, valid email format, unique.
- o Password → At least 8 characters, must include uppercase, lowercase, numbers, and special characters.
- o Confirm Password → Must exactly match the password.

2. Validation Rules:

- o Frontend validation (instant feedback before form submission).
- o Backend validation (secure check on server-side).
- o Display meaningful error messages for incorrect input.

3. Database Integration:

- o Store validated user details securely.
- o Reject duplicates (username or email).

4. User Feedback:

- o Display success message on successful registration.
- o Clear error messages when validation fails.

5. Wireframes / API Endpoint List

Wireframe (Basic UI Layout):

Username: [_____] |
| Email: [_____] |
| Password: [_____] |
| Confirm Pass: [_____] |
| [Register] |

API Endpoint List:

- POST /api/register → Registers a new user after validating inputs.
- POST /api/login → Allows registered users to log in with their credentials.
- GET /api/users/:id → Retrieves specific user details (admin use).
- DELETE /api/users/:id → Removes a user account if required (admin use).

6. Acceptance Criteria

1. Field Validation:

- o Registration fails if required fields are missing.
- o Email must be in a valid format (e.g., “user@example.com”).
- o Username must be alphanumeric and at least 3 characters long.

2. Password Rules:

- o Must contain a minimum of 8 characters.
- o Must include uppercase, lowercase, digits, and special symbols.
- o Confirm Password must match Password.

3. Uniqueness:

- o Duplicate usernames or emails are not allowed.
- o Attempt to register with an existing email should return an error.

4. System Response:

- o Success message displayed when registration completes.
- o Error messages are clear, user-friendly, and specific to the problem.

5. Database Handling:

- o Valid user details are securely stored.
- o No sensitive information (like passwords) is stored in plain text (use hashing).

6. Usability:

- o Registration form is simple, intuitive, and mobile-friendly.
- o Users are guided with proper hints (e.g., “Password must be at least 8 characters”).